

Infrastructure events

Infrastructure events are log messages that helpful for incident management to help answer the question **what changes happened leading up to the event?**

- **View events for Production**
- **View events for Staging**

We log events for the following actions:

- All infrastructure and deployment pipelines for the staging (**gstg**) and production (**gprd**) environments.
 - <https://ops.gitlab.net/gitlab-cookbooks/chef-repo/>
 - <https://ops.gitlab.net/gitlab-com/gitlab-com-infrastructure/>
 - <https://gitlab.com/gitlab-com/gl-infra/k8s-workloads>
- Chatops commands for Feature Flags and Canary
- PagerDuty events via a webhook listener <https://gitlab.com/gitlab-com/gl-infra/pd-event-logger>

Note that we do not currently differentiate between successful and failed deployments, tracked in <https://gitlab.com/gitlab-com/gl-infra/reliability/-/issues/12685>. Other environments like **pre**, **release**, etc. are not included because they are not part of incident management.

There are two Elasticsearch indexes that are used for events, **events-gstg** and **events-gprd**. These indexes are both configured in the non-prod Elasticsearch cluster `nonprod-log.gitlab.net` so that we are not tied to the availability of the production for event records.

CI Variables

There is a dedicated user **events** in the nonprod cluster for sending events in the 1Password production vault named “User for sending infra events to Elasticsearch”.

In CI, use the variable named `$ES_NONPROD_EVENTS_URL` for sending events with `curl`.

Fields for events

The following fields are recommended for events:

name	type
time	string
type	string
message	string
env	string
stage	string

name	type
username	string
source	string
diff_url	string

- **message:** Free-form text describing the event
- **env:** Either `gprd` or `gstg`.
- **stage:** Either `main` or `cny`
- **username:** GitLab username if available, if unknown use `unknown` as the value.
- **type:** The type of event, for example: `deployment`, `configuration`, `alert`, etc.
- **diff_url:** optional HTTP link, if a list of changes are available.
- **source:** optional source, may be a URL to a pipeline or job or free-form text

Sending events from CI

The following snippet can be used to create shell function that will send events from CI

```
.sendEvent:
- &sendEvent
|
sendEvent() {
    command -v curl >/dev/null 2>&1 || \
        { echo >&2 "sending events requires curl but it's not installed."; exit 1; }
    MSG="$1"
    TYPE="${2:-configuration}"
    ENV="${3:-gprd}"
    TS=$(date -u +%s000)
    USERNAME="${GITLAB_USER_LOGIN:-unknown}"
    SOURCE="${CI_JOB_URL:-unknown}"
    DATA="
        {
            \"time\": \"$TS\",
            \"type\": \"$TYPE\",
            \"message\": \"$MSG\",
            \"env\": \"$ENV\",
            \"username\": \"$USERNAME\",
            \"source\": \"$SOURCE\"
        }
    "
    echo "Sending event: \"$MSG\""
    curl -s -X POST "$ES_NONPROD_EVENTS_URL/events-$ENV/_doc" -H 'Content-Type: applicat
}
```

Then in a new or existing `before_script` section:

```
before_script:  
  - *sendEvent
```

And call the shell function from the `script:` section:

```
script:  
  - ...  
  - sendEvent "Starting an event"  
  - ...  
  - sendEvent "Finishing an event"
```